Online Examination Report

Date:	25.08.2014
Attempts:	1
Examinee:	Namukwaya, Lilian
Date of birth:	12/11/1990
Subject:	020-AGK
Result:	47,49 %
Time used:	00:01:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	2876	21.	AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
2	2464	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
3	2395	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
4	597	21.2.3.1	Design and construction	0,00	1,00
5	61	21.3.2	Hydraulic systems	0,00	1,00
6	1233	21.3.2	Hydraulic systems	1,00	1,00
7	2297	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
8	2919	21.8	FUEL SYSTEM	1,00	1,00
9	1859	21.8.2.2	Design, operation, system components, etc	1,00	1,00
10	2309	21.9.1.3	Alternating Current	0,00	1,00
11	1568	21.9.1.7	Circuit breakers	1,00	1,00
12	96	21.9.3.1	DC Generation	0,00	1,00
13	1013	21.9.3.2	AC Generation	0,00	1,00
14	1647	21.10.7	Ignition circuits	0,00	1,00
15	1579	21.10.10.1	Performance	1,00	1,00
16	2470	21.10.10.1	Performance	0,00	1,00
17	181	21.11.1.1	Basic generation of thrust, the thrust formula	0,00	1,00
18	2090	21.11.2.4	Turbine	1,00	1,00
19	956	21.12.2	Fire protection systems	0,00	1,00
20	2371	21.13	OXYGEN SYSTEMS	0,00	1,00
21	4522	22.	INSTRUMENTATION	1,00	1,00
22	3102	22.	INSTRUMENTATION	0,00	1,00
23	2636	22.	INSTRUMENTATION	1,00	1,00
24	3107	22.	INSTRUMENTATION	0,00	1,00
25	2806	22.	INSTRUMENTATION	1,00	1,00
26	4127	22.	INSTRUMENTATION	1,00	1,00
27	2815	22.	INSTRUMENTATION	1,00	1,00
28	2809	22.	INSTRUMENTATION	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter		Points	Max. Points
29	290	22.2.1	Pressure measurement		0,00	1,00
30	237	22.2.4	Altimeter		0,00	1,00
31	107	22.2.6	Airspeed indicator (ASI)		0,00	1,00
32	1158	22.2.6	Airspeed indicator (ASI)		0,00	1,00
33	1759	22.3.3	Direct Reading Magnetic Compass		0,00	1,00
34	1607	22.4.1	Gyroscope: basic principles		1,00	1,00
35	1711	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator		1,00	1,00
36	1501	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator		0,00	1,00
37	1461	22.4.2	Rate of turn and slip indicator/Turn Co-ordinator		1,00	1,00
38	884	22.6.3	Flight Director : design and operation.		0,00	1,00
39	574	22.6.3	Flight Director : design and operation.		1,00	1,00
40	1770	22.6.3	Flight Director : design and operation.		1,00	1,00
				Total:48%	**19,00**	**40,00**

Attachment 2: List of result by topic

Licence

Date: 25.08.2014

Examinee: Namukwaya, Lilian

Location: Uganda Civil Aviation Authority

Your Result: 19,00 from 40,00 Points (47,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	8,00	20,00	40,00
21.2. AIRFRAME	3	1,00	3,00	33,33
21.3. HYDRAULICS	2	1,00	2,00	50,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	1	0,00	1,00	0,00
21.8. FUEL SYSTEM	2	2,00	2,00	100,00
21.9. ELECTRICS	4	1,00	4,00	25,00
21.10. PISTON ENGINES	3	1,00	3,00	33,33
21.11. TURBINE ENGINES	2	1,00	2,00	50,00
21.12. PROTECTION AND DETECTION SYSTEMS	1	0,00	1,00	0,00
21.13. OXYGEN SYSTEMS	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	11,00	20,00	55,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	4	0,00	4,00	0,00
22.3. MAGNETISM - DIRECT READING COMPASS AND FLUX VALVE	1	0,00	1,00	0,00
22.4. GYROSCOPIC INSTRUMENTS	4	3,00	4,00	75,00
22.6. AEROPLANE : AUTOMATIC FLIGHT CONTROL SYSTEMS	3	2,00	3,00	66,67
Total	**40**	**19,00**	**40,00**	**47,50**

1

What should help prevent aircraft induced oscillation on a gyroplane?

- [x] **Adding a horizontal stabilizer.**
- [ ] **Lowering the center of gravity below the thrust line.**
- [ ] **Increasing cyclic control sensitivity.**

ID:	2876(LTS:222981, LTM:1107)	Max. Points	1,00
Syllabus reference:	21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	Points	1,00

2

What should a pilot do to maintain 'best range' airplane performance when a tailwind is encountered?

- ☐ **Increase speed.**
- ☒ **Decrease speed.**
- ☐ **Maintain speed.**

ID:	2464(LTS:222906, LTM:1043)	Max. Points	1,00
Syllabus reference:	21.2.3 Aeroplane: Wings, tail and control surfaces	Points	1,00

3

When are outboard ailerons normally used?

- ☒ **High-speed flight only.**
- ☐ **Low-speed and high-speed flight.**
- ☐ **Low-speed flight only.**

ID:	2395(LTS:222815, LTM:974)	Max. Points	1,00
Syllabus reference:	21.2.3 Aeroplane: Wings, tail and control surfaces	Points	0,00

4

Among the different types of aircraft structures, the shell structures efficiently transmit the: 1. normal bending stresses 2. tangent bending stresses 3. torsional moment 4. shear stresses The combination regrouping all the correct statements is :

- ☐ **1, 2, 3**
- ☒ **1, 2, 4**
- ☐ **1, 3, 4**
- ☐ **2, 3, 4**

ID:	597(LTS:222375, LTM:131)	Max. Points	1,00
Syllabus reference:	21.2.3.1 Design and construction	Points	0,00

5

Large transport aeroplane hydraulic systems usually operate with a system pressure of approximately:

- ☐ **3000 psi**
- ☒ **2000 psi**
- ☐ **1000 psi**
- ☐ **4000 psi**

ID:	61(LTS:222258, LTM:15)	Max. Points	1,00
Syllabus reference:	21.3.2 Hydraulic systems	Points	0,00

6

The component that transforms the hydraulic pressure into a linear motion is called ...

- [x] **An actuator or jack.**
- [ ] **A Pressure regulator.**
- [ ] **An accumulator.**
- [ ] **A hydraulic pump.**

ID:	1233(LTS:222524, LTM:280)	Max. Points	1,00
Syllabus reference:	21.3.2 Hydraulic systems	Points	1,00

7

Assuming cabin differential pressure has attained the required value in normal flight conditions, if flight altitude is maintained:

- ☐ **the outflow valves will move to the fully closed position.**
- ☒ **the outflow valves will move to the fully open position.**
- ☐ **the pressurisation system ceases to function until leakage reduces the pressure.**
- ☐ **a constant mass air flow is permitted through the cabin.**

ID:	2297(LTS:222803, LTM:940)	Max. Points	1,00
Syllabus reference:	21.6.3 Aeroplane: Pressurisation and air conditioning system	Points	0,00

8

The purpose of the preheating coil used in hot air balloons is to

- ☐ prevent ice from forming in the fuel lines.
- ☒ vaporize the fuel for more efficient burner operation.
- ☐ warm the fuel tanks for more efficient fuel flow.

ID:	2919(LTS:222951, LTM:1121)	Max. Points	1,00
Syllabus reference:	21.8 FUEL SYSTEM	Points	1,00

9

Fuel pumps submerged in the fuel tanks of a large aeroplane are:

- ☐ ***low pressure variable swash plate pumps.***
- ☐ ***high pressure swash plate pumps.***
- ☐ ***centrifugal high pressure pumps.***
- ☒ ***centrifugal low pressure type pumps.***

ID:	1859(LTS:222711, LTM:459)	Max. Points	1,00
Syllabus reference:	21.8.2.2 Design, operation, system components, etc	Points	1,00

10 *In an alternator rotor coil you can find :*

- ☒ **Three-phase AC.**
- ☐ **Only induced current.**
- ☐ **AC.**
- ☐ **DC.**

ID: 2309(LTS:222807, LTM:944)
Syllabus reference: 21.9.1.3 Alternating Current

Max. Points 1,00
Points 0,00

11

Assuming the initiating cause is removed, which of these statements about resetting are correct or incorrect?
I. A fuse is not resettable
II. A circuit breaker is resettable.

- [x] *I is correct, II is correct.*
- [ ] *I is correct, II is incorrect.*
- [ ] *I is incorrect, II is incorrect.*
- [ ] *I is incorrect, II is correct.*

ID:	1568(LTS:222610, LTM:367)	Max. Points	1,00
Syllabus reference:	21.9.1.7 Circuit breakers	Points	1,00

12

The frequency of the current provided by an alternator depends on...

- ☐ *its rotation speed.*
- ☐ *its load.*
- ☐ *its phase balance.*
- ☐ *the strength of the excitation current.*

ID:	96(LTS:222265, LTM:21)	Max. Points	1,00
Syllabus reference:	21.9.3.1 DC Generation	Points	0,00

13

In order to produce an alternating voltage of 400 Hz, the number of poles required in an AC generator running at 6000 rpm is:

- ☐ **4**
- ☒ **12**
- ☐ **24**
- ☐ **8**

ID:	1013(LTS:222446, LTM:222)	Max. Points	1,00
Syllabus reference:	21.9.3.2 AC Generation	Points	0,00

14

If an engine fails to stop with the magneto switch in OFF position, the cause may be:

- ☐ *switch wire grounded*
- ☒ *fouled spark plugs*
- ☐ *excessive carbon formation in cylinder head.*
- ☐ *defective condenser*

ID:	1647(LTS:222643, LTM:390)	Max. Points	1,00
Syllabus reference:	21.10.7 Ignition circuits	Points	0,00

15

During a climb in a standard atmosphere with constant MAP and RPM indications and at a constant mixture setting, the power output of a piston engine:

- ☒ ***increases.***
- ☐ ***only stays constant if the propeller lever is pushed.***
- ☐ ***stays constant.***
- ☐ ***decreases.***

ID: 1579(LTS:222615, LTM:372)
Syllabus reference: 21.10.10.1 Performance

Max. Points 1,00
Points 1,00

16

How does V(NE) speed vary with altitude?

- ☐ **Varies inversely with altitude.**
- ☐ **Remains the same at all altitudes.**
- ☒ **Varies directly with altitude.**

ID:	2470(LTS:222912, LTM:1049)	Max. Points	1,00
Syllabus reference:	21.10.10.1 Performance	Points	0,00

17

The fan in a high by-pass ratio turbo-jet engine produces:

- ☐ none of the thrust.
- ☐ the greater part of the thrust.
- ☒ the lesser part of the thrust.
- ☐ half the thrust.

ID:	181(LTS:222282, LTM:38)	Max. Points	1,00
Syllabus reference:	21.11.1.1 Basic generation of thrust, the thrust formula	Points	0,00

18

The primary reason for a limitation being imposed on the temperature of gas flow is to :

- [x] **ensure that the maximum acceptable temperature at the turbine blades is not exceeded.**
- [ ] **prevent overheating and subsequent creep of the nozzle**
- [ ] **prevent damage to the jet pipe from overheating.**
- [ ] **ensure that the maximum acceptable temperature within the combustion chamber is not exceeded.**

ID:	2090(LTS:222751, LTM:888)	Max. Points	1,00
Syllabus reference:	21.11.2.4 Turbine	Points	1,00

19

A smoke mask is a :

- ☐ mask with flow on request and covers the whole face.
- ☐ mask with flow on request and covers only the nose and the mouth.
- ☒ continuous flow mask and covers the whole face.
- ☐ continuous flow mask and covers only the nose and the mouth.

ID:	956(LTS:222461, LTM:208)	Max. Points	1,00
Syllabus reference:	21.12.2 Fire protection systems	Points	0,00

20

What is breathed in when using a passenger oxygen mask?

- ☐ **A mixture of oxygen and freon gas.**
- ☐ **Cabin air and oxygen.**
- ☐ **Cabin air and oxygen or 100% oxygen.**
- ☒ **100% oxygen.**

ID:	2371(LTS:222812, LTM:971)	Max. Points	1,00
Syllabus reference:	21.13 OXYGEN SYSTEMS	Points	0,00

21

Which factors must be recorded by the approved flight recorder?

- ☐ **Time, true altitude, calibrated airspeed, vertical speed, and heading.**
- ☒ **Airspeed, time, altitude, vertical acceleration, and heading.**
- ☐ **Elapsed time, airspeed, altitude, vertical acceleration, and magnetic course.**

ID:	4522(LTS:223044, LTM:1181)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

22

Which may lead to a power push-over in a gyroplane?

☐ **Rotor force is removed.**

☒ **Decreasing power too quickly.**

☐ **Low speed.**

ID:	3102(LTS:222995, LTM:1132)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

23

A gyroplane will have the greatest tendency to roll during

- ☐ **descending flight in which forward airspeed decreases.**
- ☐ **climbing flight in which forward airspeed decreases.**
- ☒ **horizontal flight at high speed.**

ID:	2636(LTS:222937, LTM:1074)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

24

What are the indications of the pulsating VASI?

- ☒ **High - pulsing white, on glidepath - green, low - pulsing red.**
- ☐ **High - pulsing white, on course and on glidepath - steady white, off course but on glidepath - pulsing white and red; low - pulsing red.**
- ☐ **High - pulsing white, on glidepath - steady white, slightly below glide slope steady red, low - pulsing red.**

ID:	3107(LTS:223006, LTM:1133)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	0,00

25

Which statement is true about magnetic deviation of a compass?

- [x] **Deviation varies for different headings of the same aircraft.**
- [ ] **Deviation is the same for all aircraft in the same locality.**
- [ ] **Deviation is different in a given aircraft in different localities.**

ID:	2806(LTS:222969, LTM:1095)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

26

(Refer to Figure 85.) What route should you take if cleared for the Washoe Two Departure and your assigned route is V6?

See Attachments:

1. instrument_85.jpg	figure 85	(Attachment No.1)

- ☐ **Climb on the LOC south course to cross WAGGE at 9,000, turn left and fly direct to FMG VORTAC and cross at or above 10,000, and proceed on FMG R-241.**
- ☒ **Climb on the LOC south course to WAGGE where you will be vectored to V6.**
- ☐ **Climb on the LOC south course to WAGGE, turn left and fly direct to FMG VORTAC. If at 10,000 turn left and proceed on FMG R-241; if not at 10,000 enter depicted holding pattern and climb to 10,000 before proceeding on FMG R-241.**

ID:	4127(LTS:223034, LTM:1171)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

27

If a pitot tube is clogged, which instrument would be affected?

☐ **Altimeter.**

☐ **Vertical speed indicator.**

☒ **Airspeed indicator.**

ID: 2815(LTS:222971, LTM:1097)
Syllabus reference: 22. INSTRUMENTATION

Max. Points 1,00
Points 1,00

28

Which instrument would be affected by excessively low pressure in the airplane's vacuum system?

- ☐ Airspeed indicator.
- ☐ Pressure altimeter.
- ☒ Heading indicator.

ID:	2809(LTS:222970, LTM:1096)	Max. Points	1,00
Syllabus reference:	22. INSTRUMENTATION	Points	1,00

29

The location of the static vent which could provide the most accurate measurement of static pressure under variable flight conditions is:

- ☒ **At the Pitot head which encounters relatively undisturbed air.**
- ☐ **One on each side of the aircraft where the system will compensate for variation of aircraft attitude.**
- ☐ **In the cockpit where it is not influenced by a variable angle of attack.**

ID:	290(LTS:222306, LTM:62)	Max. Points	1,00
Syllabus reference:	22.2.1 Pressure measurement	Points	0,00

30

An aircraft leaves FL 160 for an approach and landing at an airfield. The pilot will set QNH at the:

☐ **Transition level.**

☐ **Transition Layer.**

☒ **Transition Altitude.**

ID:	237(LTS:222294, LTM:50)	Max. Points	1,00
Syllabus reference:	22.2.4 Altimeter	Points	0,00

31

An aircraft is maintaining FL 120 in cloud. The ASI reading falls to zero. The most probable cause is:

- ☐ **Static vent blocked by ice.**
- ☐ **ASI malfunction.**
- ☒ **Pitot head and static vent blocked by ice.**

ID:	107(LTS:222268, LTM:24)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

32

The limits of the green scale of an airspeed indicator are:

- ☐ ***VS1 and VNO.***
- ☐ ***VS0 and VNE.***
- ☒ ***VS1 and VNE.***
- ☐ ***VS1 and VMO.***

ID:	1158(LTS:222505, LTM:261)	Max. Points	1,00
Syllabus reference:	22.2.6 Airspeed indicator (ASI)	Points	0,00

33

In a steep turn, the northerly turning error on a direct reading magnetic compass on the northern hemisphere is:

- ☐ *equal to 180° on a 090° heading in a right turn.*
- ☐ *none on a 090° heading in a right turn.*
- ☒ *equal to 180° on a 270° heading in a right turn.*
- ☐ *none on a 270° heading in a left turn.*

ID:	1759(LTS:222670, LTM:427)	Max. Points	1,00
Syllabus reference:	22.3.3 Direct Reading Magnetic Compass	Points	0,00

34

An Artificial Horizon employs a;

- [x] **Earth gyro**
- [ ] **Rate gyro**
- [ ] **Tied gyro**

ID:	1607(LTS:222620, LTM:377)	Max. Points	1,00
Syllabus reference:	22.4.1 Gyroscope: basic principles	Points	1,00

35

During a stabilised climbing turn at a constant rate, the instruments which indicate the correct pitch and bank are the:

- ☒ **attitude indicator and turn-and-slip indicator.**
- ☐ **altimeter and turn-and-slip indicator;**
- ☐ **vertical-speed indicator and turn-and-slip indicator;**

ID:	1711(LTS:222662, LTM:410)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	1,00

36

What indications should you get from the Turn and Slip indicator during taxi?

- ☒ **The ball deflects opposite to the turn and the needle remains central.**
- ☐ **The ball moves opposite to the turn and the needle deflects in the direction of the turn.**
- ☐ **The needle and ball should move freely in the direction of the turn.**

ID:	1501(LTS:222592, LTM:349)	Max. Points	1,00
Syllabus reference:	22.4.2 Rate of turn and slip indicator/Turn Co-ordinator	Points	0,00

37

What is the approximate angle of bank for a rate one turn at 110 knots?

- ☐ **30 degrees**
- ☐ **25 degrees**
- ☒ **18 degrees**

ID: 1461(LTS:222582, LTM:339)
Syllabus reference: 22.4.2 Rate of turn and slip indicator/Turn Co-ordinator

Max. Points 1,00
Points 1,00

38

The parameters taken into account by the flight director computer in the altitude hold mode (ALT HOLD) are:

1 - altitude deviation
2 - engine rpm
3 - ground speed
4 - pitch attitude

The combination that regroups all of the correct statements is:

☒ ***1, 2.***

☐ ***1, 3.***

☐ ***1, 4.***

☐ ***1.***

ID:	884(LTS:222433, LTM:189)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	0,00

39

The vertical command bar of a flight director:

1 - repeats the position information given by the ILS in the horizontal plane
2 - repeats the position information given by the ILS in the vertical plane
3 - gives information about the direction and the amplitude of the corrections to be applied on the bank of the aircraft

The combination that regroups all of the correct statements is:

- ☐ *1.*
- ☐ *1, 3.*
- ☒ *3.*
- ☐ *2, 3.*

ID:	574(LTS:222368, LTM:124)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	1,00

40

The flight director indicates the:

- ☐ ***optimum trajectory at the moment it is entered to reach a selected radial.***
- ☐ ***trajectory permitting reaching a selected radial over a minimum distance.***
- ☒ ***optimum instantaneous trajectory to reach selected radial.***
- ☐ ***trajectory permitting reaching a selected radial in minimum time.***

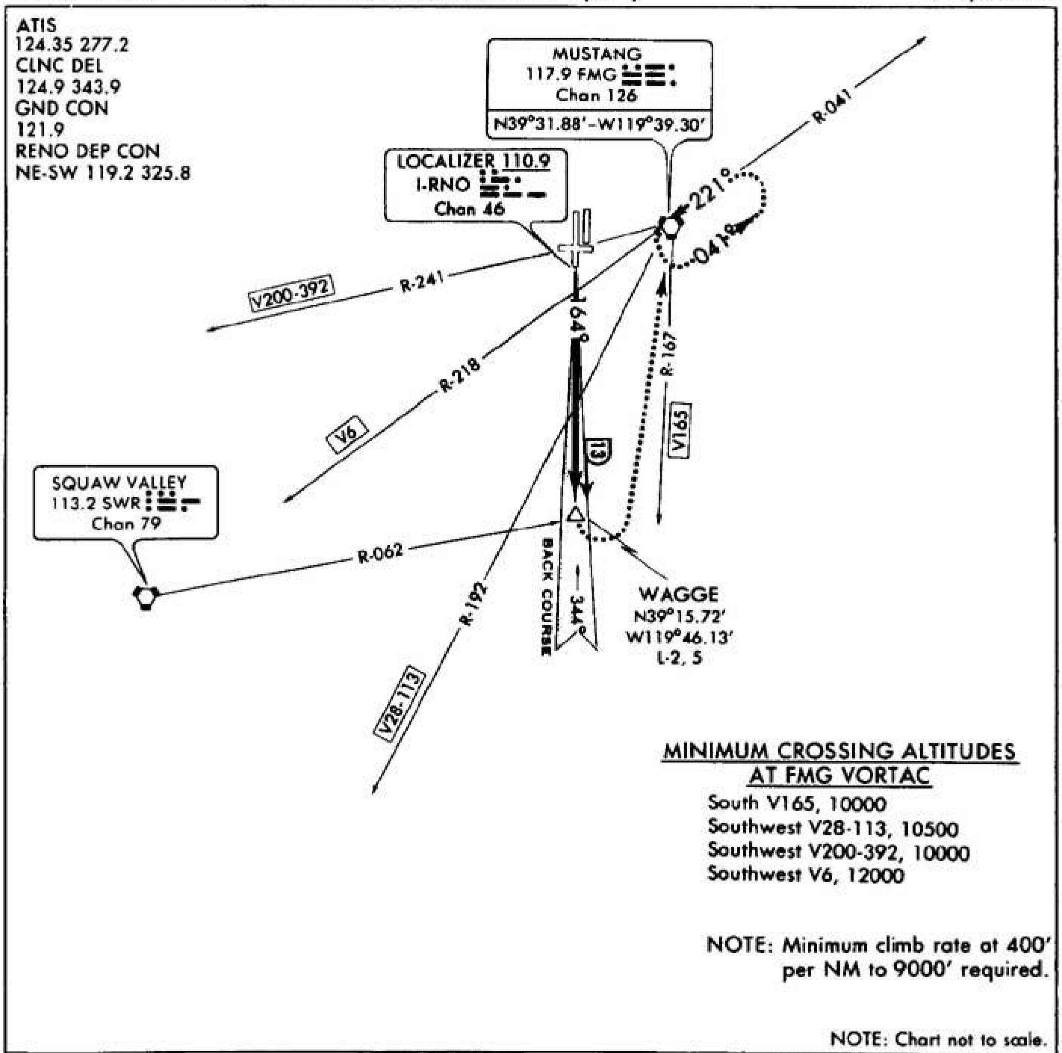
ID:	1770(LTS:222673, LTM:430)	Max. Points	1,00
Syllabus reference:	22.6.3 Flight Director : design and operation.	Points	1,00

Attachment No. 1

Questions 26
figure 85
instrument_85.jpg

(VECTOR) (WASH2.WAGGE) 91094
WASHOE TWO DEPARTURE SL-346 (FAA)

RENO CANNON INTL
RENO, NEVADA



DEPARTURE ROUTE DESCRIPTION

<u>TAKE-OFF RUNWAYS 16L/R:</u> Climb via I-RNO Localizer south course to WAGGE INT then via radar vectors to assigned route.

<u>LOST COMMUNICATIONS:</u> If not in contact with departure control within one minute after takeoff, or if communications are lost before reaching 9000', continue climb via I-RNO localizer south course to WAGGE INT, turn left, proceed direct FMG VORTAC. Cross FMG VORTAC at or above MCA, thence via assigned route or climb in holding pattern northeast on FMG R-041, left turns to cross FMG VORTAC at or above MCA for assigned route.

WASHOE TWO DEPARTURE
(VECTOR) (WASH2.WAGGE)

RENO, NEVADA
RENO CANNON INTL

Figure 85. WASHOE TWO DEPARTURE © ASA